

Lec 04: Performance modeling of computer networks

Computer Networks (CSE232) Section A

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These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
6th edition, Jim Kurose, Keith Ross, Pearson, 2017
2. *Course notes by Prof. Mythili Vutukuru, IIT Bombay*
https://www.cse.iitb.ac.in/~mythili/teaching/cs641_autumn2015/index.html



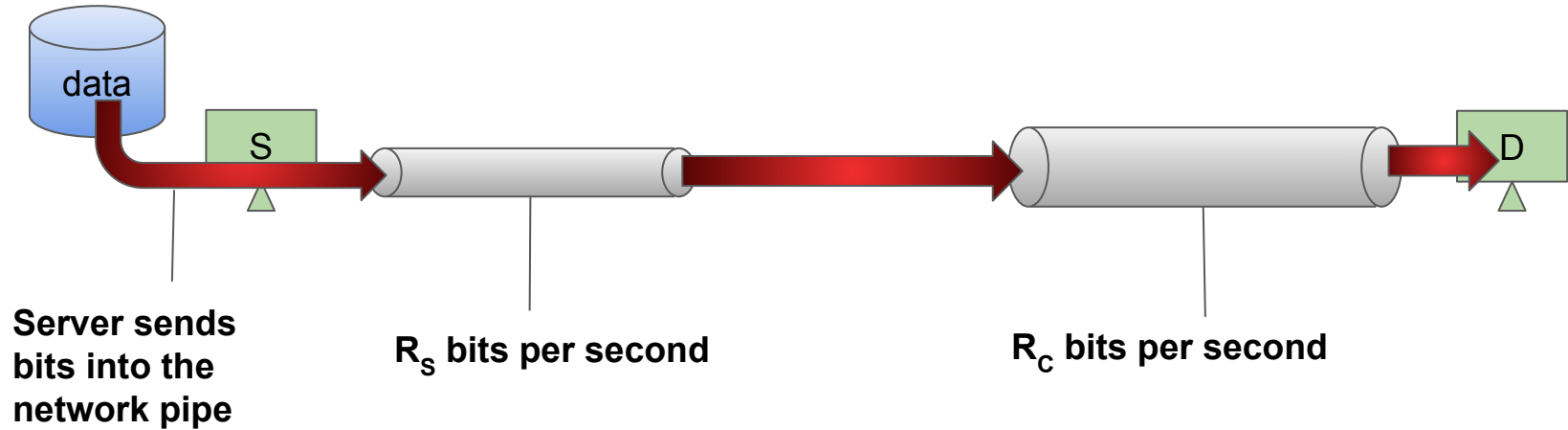
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DELHI

Modeling the performance of computer networks

Two main attributes to measure network performance

- **Throughput**
 - Rate at which bits are transferred between the sender and receiver
- **Delay**
 - Time taken for a bit to reach from the sender to receiver

Throughput



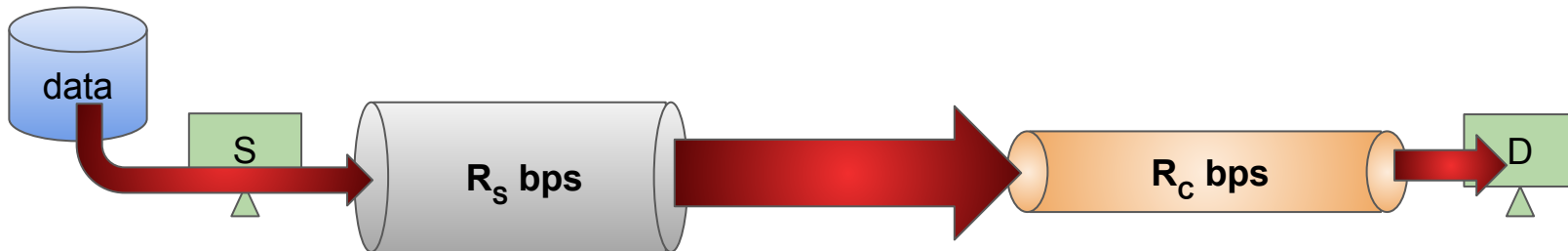
Throughput: Rate at which bits are transferred between the sender and receiver (bits per second)

Throughput (contd.)

$R_s < R_c$ What is the average end-end throughput?

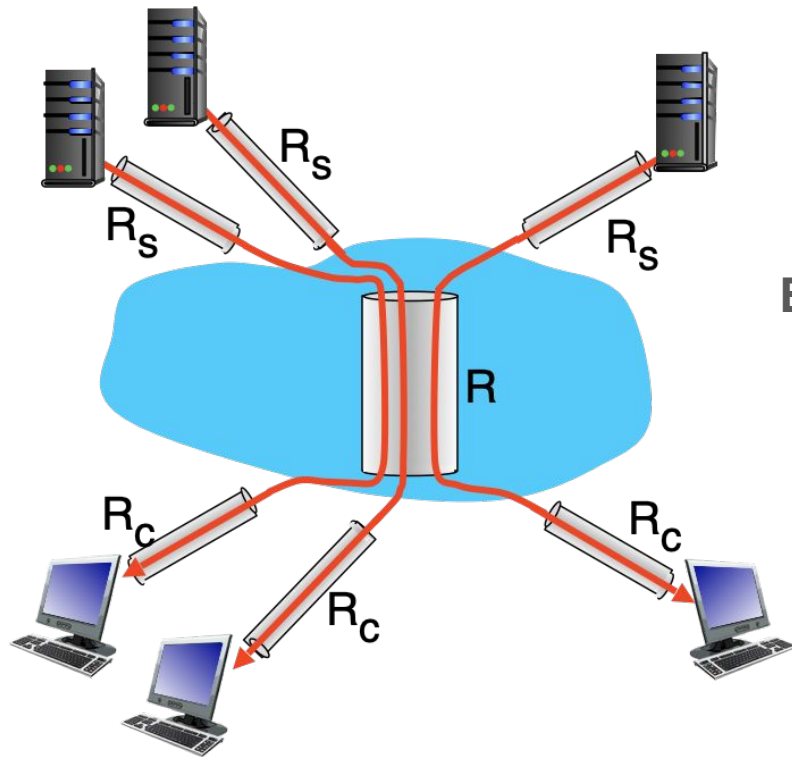


$R_s > R_c$ What is the average end-end throughput?



Bottleneck link on the path constraints end-end throughput

Throughput: Internet scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

End-to-end throughput per connection?

- $\min(R_S, R_C, R/10)$
- Typical bottleneck, R_C or R_S

Average throughput **NOT ALWAYS** = Bottleneck bandwidth

Reasons

- Protocol may add delays
 - Sender has to wait for an ACK from receiver (flow control)
- Packets losses may lead to retransmissions
 - You may not always send useful data
 - GOODPUT: Number of useful bits per sec

**Instantaneous throughput may be the bottleneck rate;
Average throughput may be lower!**

Throughput calculation

$$\text{Throughput} = \text{Amount of data sent} / \text{Time taken}$$

Example:

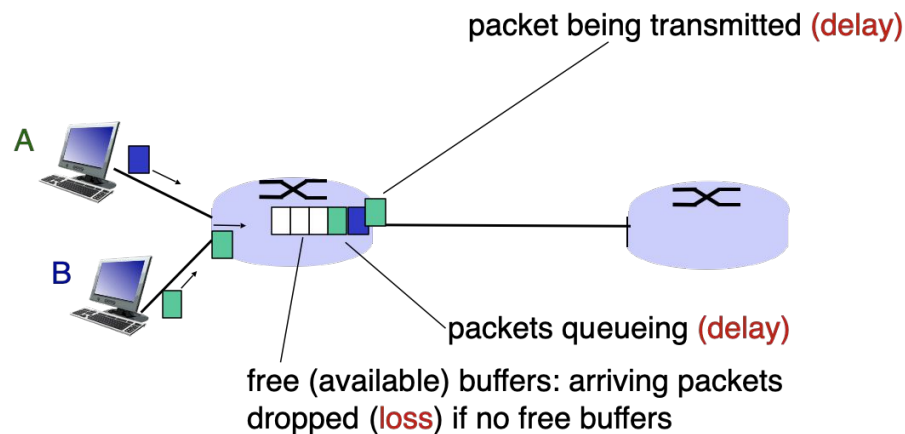
A file of size `L` bits takes `d` seconds to be transferred.

Average throughput = L/d bps

Problem

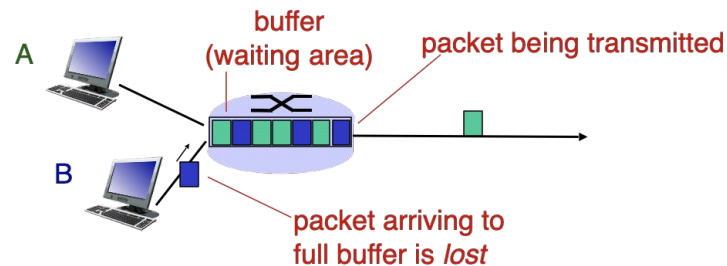
1. Consider a 125KB file that needs to be sent through a network path. The bottleneck bandwidth is 1 Mbps. Suppose the sender continuously sends data at the bottleneck rate, and no packets are lost, no retransmissions. How long does it take to send the file?
2. Now, suppose one way delay between the sender and receiver is 20 ms. Suppose sender needs to wait for an ACK after sending every 1KB packet. Assume ACK takes 20 ms to come back. What is the time required to send 1KB packet? What is the throughput?

Packet switching: How do packet delay or packet losses occur?



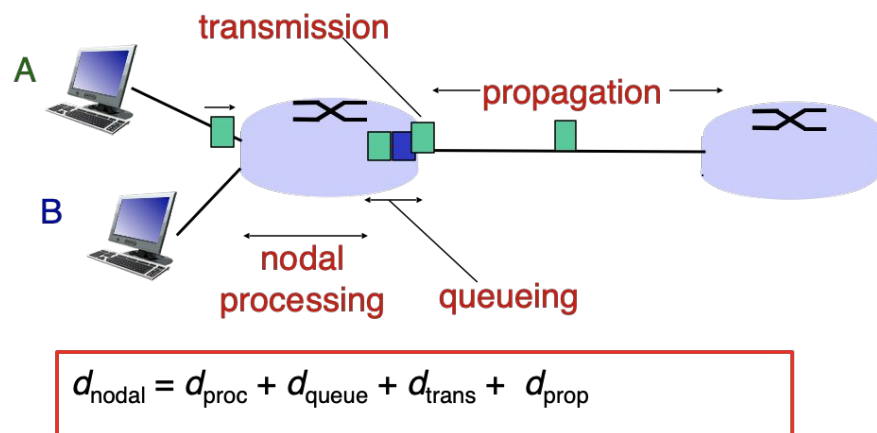
Packet arriving rate > Output link's capacity (temporarily)

- Incoming packet stored in queue (router's buffers) waiting for its turn



- Finite buffer capacity at the router
- Packet arrives when queue is full; packet is dropped
- Lost packet may or may not be retransmitted
 - Retransmission by previous node
 - Retransmission by source

Sources of packet delay



- **Transmission delay (d_{trans})**

- Time required to place the entire packet over the output link
- Depends on link capacity (bandwidth)

- **Propagation delay (d_{prop})**

- Time required for one bit to propagate from one end to another
- Depends on transmission medium characteristics

- **Nodal processing delay (d_{proc})**

- Check bit errors
- Determine output link
- Depends on node specs

- **Queueing delay (d_{queue})**

- Waiting time at output link for congestion
- Depends on congestion level

Sources of packet delay (contd.)

- Transmission delay

- Link data rate = R bits/sec
- Packet size = L bits
- Transmission delay, $d_{\text{trans}} = L/R$ seconds
- Example: Packet size, $L = 1000$ bits, Link data rate = 10^9 bps; $d_{\text{trans}} = 1$ us

- Propagation delay

- Length of the link / distance travelled = l meters
- Speed at which signal travels = s meters/sec
- Propagation delay, $d_{\text{prop}} = l/s$ seconds
- Example: Speed of light, $s = 3 \times 10^8$ m/s; $l = 300$ m; $d_{\text{prop}} = 1$ us

Relation between the link utilization & propagation/transmission delays

$$d_{\text{prop}} < d_{\text{trans}}$$

- The first bit reaches the receiver before the sender finishes transmission
- The data pipe is FULL

$$d_{\text{prop}} > d_{\text{trans}}$$

- Sender finishes transmission; receiver has not received even the first bit
- Typical scenario in case of long distance links

- **Limiting factor: How fast the link is?**
- **Design packet size for better network utilization (pipe should be FULL)**